

Probability

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1 Probability

1.1 Combinatorics

The theory of counting is known as **combinatorics**. It deals with counting the number of ways that an event can occur.

The main approach taken involves simplifying the solution by decomposing the problem into smaller parts.

1.1.1 The product rule

If a procedure can be broken down into a sequence of two tasks, and there are n_1 ways to do the first task, and n_2 ways to do the second task, then the total number of ways to complete the procedure is $n_1 n_2$.

This can be generalised to:

- If a sequence of tasks $T_1, T_2, \dots, T_m$ can be done in $n_1, n_2, \dots, n_m$ ways respectively, and every task arrives after the occurrence of the previous task, then there are $n_1 \cdot n_2 \cdot \dots \cdot n_m$ ways to perform the tasks.

1.1.2 The sum rule

If a procedure requires only one of two independent tasks to be completed, and there are n_1 ways to do the first task, and n_2 ways to do the second task, then the total number of ways to complete the procedure is $n_1 + n_2$.

This can be generalised to:

- If a sequence of tasks $T_1, T_2, \dots, T_m$ can be done in $n_1, n_2, \dots, n_m$ ways respectively (with no tasks being performed simultaneously), then there are $n_1 + n_2 + \dots + n_m$ ways to perform one of the tasks.

The product rule is used when the sub-tasks of an event are dependent on one another (each of them is performed); the sum rule is used when they are independent (only one of them is performed)

1.1.3 Inclusion-exclusion principle

If two tasks can be performed at the same time, there is a possibility of overcounting, since some ways may be counted twice. To prevent this:

- Add the number of ways to perform each task
- Then subtract the number of ways to do both tasks

1.1.4 Permutations

A permutation is an ordered arrangement of objects. An ordered arrangement of r objects is called an r -permutation.

- ABC, ACB, BAC, BCA, CAB and CBA are all the unique 3-permutations of the set $\{A, B, C\}$

The number of r -permutations for a set of n distinct elements is given by:

$$P(n, r) = n \times (n - 1) \times (n - 2) \times \cdots \times (n - r + 1)$$

This can be expressed more simply in factorial form:

$$P(n, r) = \frac{n!}{(n - r)!}$$

This formula assumes all objects in the set are distinct. If the set contains n objects, of which n_1 are of one type (indistinguishable from each other), n_2 are of a second type, n_k are of a k_{th} type, so that $n = n_1 + n_2 + \cdots + n_k$, then the number of distinct permutations is given by:

$$P(n; n_1, n_2, \cdots, n_k) = \frac{n!}{n_1! n_2! \cdots n_k!}$$

1.1.5 Combinations

A combination is an unordered arrangement of objects. An unordered arrangement of r objects is called an r -combination.

- While a, b and b, a would be unique permutations, they represent the same combination.

The number of r -combinations for a set of n distinct elements is given by:

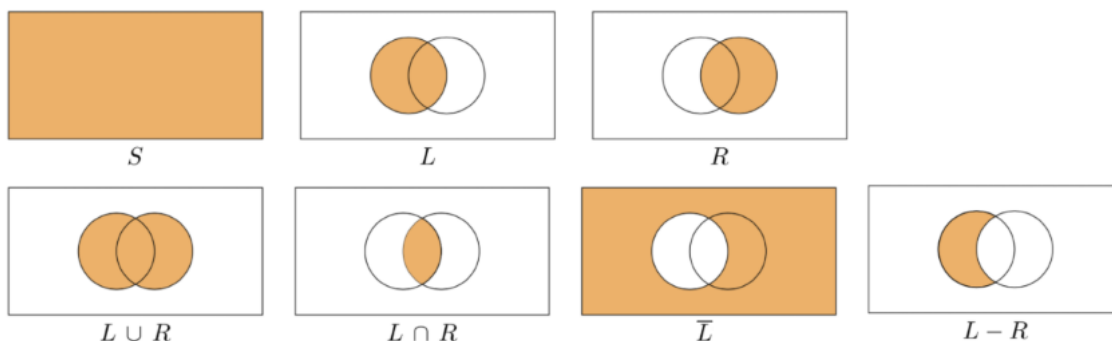
$$C(n, r) = \binom{n}{r} = \frac{n!}{(n - r)! r!}$$

1.2 Set theory

A set is a collection of unordered objects. The size, or cardinality, of a set L , $|L|$, is the number of elements in the set., is the number of elements in the set.

For two sets L and R , the basic set operations are:

- Union: $L \cup R$ - the set of elements in L or R or both
- Intersection: $L \cap R$ - the set of elements in both L and R
- Complement: $\bar{L}$ - all the elements not in L
- Set Difference/Subtract: $L - R$ - elements in L that are not in R
- Subset: $L \subseteq S$ - every element of L is also in S



1.3 Theory of probability

An *experiment* is a procedure that yields one of a given set of outcomes, e.g. flipping a coin. The *sample space* of an experiment is the set of possible outcomes, e.g. heads or tails. An *event* is any subset of the sample space, e.g. heads.

In a finite sample space S , where each outcome is equally likely, the probability of an event $E \subseteq S$ occurring is

$$P(E) = \frac{|E|}{|S|}$$

There are three axioms of probability:

- $0 \leq P(E) \leq 1$ - the probability of any event occurring is always between 0 and 1
- $P(S) = 1$ - the probability that some outcome in the sample space occurs is 1, i.e. the outcome is always in the sample space
- For disjoint (mutually exclusive) events $E_1, E_2, \dots, E_n$ (events with no intersection):

$$- P(E_1 \cup E_2 \cup \dots \cup E_n) = \sum_{i=1}^n P(E_i)$$

Other rules can be derived from these:

- Complement rule: $P(\overline{E}) = 1 - P(E)$
- Impossible event rule: $P(\emptyset) = 0$ - the probability of an event that can never occur is 0
- General addition rule: $P(E_1 \cup E_2) = P(E_1) + P(E_2) - P(E_1 \cap E_2)$
- Subset rule: If $E_1 \subseteq E_2$, then $P(E_1) \leq P(E_2)$

1.4 Relationship of events

Events can happen concurrently, after each other or because of each other (or even more complicated).

Depending on the nature of the relationship, the probability of each event can be calculated based on other events.

1.4.1 Conditional probability

Conditional probability is the probability that an event occurs given that another event has already occurred.

The probability of event A occurring given that event B has occurred (the conditional probability) is given by:

$$P(A | B) = \frac{P(A \cap B)}{P(B)}$$

Therefore, by multiplying by $P(B)$, the probability of A and B is:

$$P(A \cap B) = P(B) \times P(A | B)$$

This is called the chain rule.

1.4.2 Dependence

Two events are independent if the occurrence of one event is not dependent on the occurrence of another.

Two events A and B are independent if and only if:

$$P(A \cap B) = P(A) \times P(B)$$

Equivalently, if they are independent, the probability of A will not change given that B happens:

$$P(A | B) = P(A)$$

1.4.3 Bayes' Theorem

Bayes' theorem allows one conditional probability, $P(A | B)$, to be calculated from another, $P(B | A)$:

$$P(A | B) = \frac{P(B | A) \times P(A)}{P(B)}$$

A general form of Bayes' theorem is as follows:

- suppose $E_1, E_2, \dots, E_n$ are mutually exclusive events such that $\bigcup_{i=1}^n E_i = S$, where S is the sample space
- then, if A is any event, Bayes' Theorem states:

$$P(E_k | A) = \frac{P(E_k) \times P(A | E_k)}{\sum_{j=1}^n (P(E_j) \times P(A | E_j))}$$

1.5 Tree diagrams

A tree consists of a root, a number of branches, each leading to other branches, ending at leaf nodes. Branches are used for each possible choice.

To find the probability of a particular outcome (leaf), multiply the probabilities along the branches that lead to that outcome.

1.6 Random variables

A random variable is a function that assigns a number to each outcome in the sample space of a random experiment. It is usually denoted by a capital letter such as X or Y .

A random variable X is discrete if it has countably many values (e.g., $x_1, x_2, x_3, \dots$).

For example, suppose a fair coin is tossed twice. The sample space will be $\{HH, HT, TH, TT\}$. A random variable X can be created to count the number of heads that occur. X will assign a numerical value to each outcome in the sample space:

- HH: 2
- HT: 1

- TH: 1
- TT: 0

1.7 Probability distribution

The probability distribution of a random variable X describes the probabilities associated with each possible outcome of X .

Let X be a discrete random variable, which can assume values $x_1, x_2, \dots, x_n$. Then, these values are associated with probabilities given by

$$P(X = x_k) = f(x_k), k = 1, 2, \dots, n$$

This function f is known as the probability mass function (PMF).

1.7.1 Probability mass function

For a discrete random variable X with possible values of $x_1, x_2, \dots$, the probability mass function is a function $f : x_1, x_2, \dots \rightarrow [0, 1]$ satisfying three properties:

- $f(x_i) \geq 0$ - each probability is positive (a probability can't be negative)
- $\sum_{k=1}^{\infty} f(x_k) = 1$ - all the probabilities sum to 1
- $P(X = x_i) = f(x_i)$ - it associates a value of the random variable to the probability of that value

To compute a PMF ($P(X = x)$):

1. collect all possible outcomes where $X = x$
2. sum their probabilities
3. repeat for all x

Continuing with the example of tossing a fair coin twice:

- HH: $X = 2$, $P(X=2) = 1/4$
- HT: $X = 1$, $P(X=1) = 1/2$
- TH: $X = 1$, $P(X=1) = 1/2$
- TT: $X = 0$, $P(X=0) = 1/4$

1.7.2 Cumulative distribution function

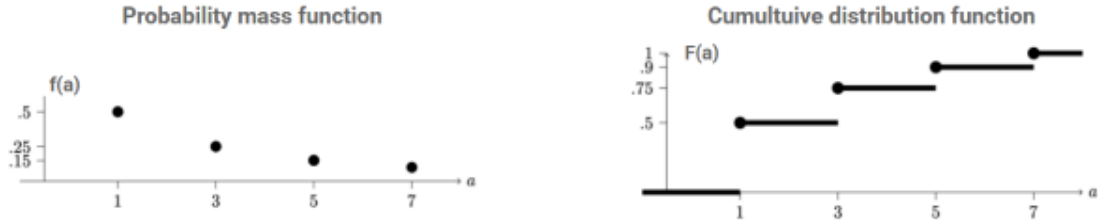
The cumulative distribution function (CDF) is a function $F : \mathbb{R} \rightarrow [0, 1]$ that gives the probability that X is less than or equal to a given value x . It is defined as follows:

$$F(X) = P(X \leq x) = \sum_{x_i \leq x} P(X = x_i) = \sum_{x_i \leq x} f(x_i), \quad \text{where } x \in \mathbb{R}$$

Note the use of F for CDF and f for PMF.

For discrete variables, PMF is used more than CDF.

All of the information contained in the PMF is also contained in the CDF. However, the CDF is properly defined for any random variable and any real number, and not only for the values X can take, unlike the PMF which is restricted to the values X can take.



1.8 Mean and variance

Mean and variance are often used to summarise a probability distribution for a random variable X .

The mean (or expected value) is the centre/middle of the probability distribution:

$$\mu = E(X) = \sum_x x f(x)$$

Variance is a measure of the dispersion/variability in the distribution:

$$\sigma^2 = V(X) = E[(X - \mu)^2] = \sum_x (x - \mu)^2 f(x) = \sum_x x^2 f(x) - \mu^2$$

Standard deviation is the square root of variance:

$$\sigma = \sqrt{\sigma^2}$$

1.9 Functions of random variables

Functions of random variables are also random variables:

- let X be a random variable and let $h : \mathbb{R} \rightarrow \mathbb{R}$ be an arbitrary function, then $Y = h(X)$ is also a random variable

Let X be a random variable with PMF f and $h : \mathbb{R} \rightarrow \mathbb{R}$ be an arbitrary function, then $Y = h(X)$ is also a random variable. If X is discrete and takes values $\{x_1, x_2, \dots\}$ then:

$$E[Y] = E[h(X)] = \sum_i h(x_i) f(x_i)$$

These laws mean that a new random variable can be formed by performing arithmetic operations on a random variable, and the expectation (mean) and variance of the new variable can be computed from the originals.

Some properties that can be derived from this information include: - $E(aX + b) = aE(X) + b$ - $E[h(X)] = \sum_x h(x)f(x)$ - $V(aX + b) = a^2V(X)$ - $V(X) = E[X^2] - (E[X])^2$

$E[X^2]$ is known as the second moment, and $E[X]$ as the first moment. The variance is also called the central second moment.

1.10 Independence of random variables

Two random variables X and Y are independent if for any sets A and B :

$$P(X \in A \cap Y \in B) \equiv P(X \in A, Y \in B) = P(X \in A) P(Y \in B)$$

For independent random variables $X_1, X_2, \dots, X_n$:

$$V(X_1 + X_2 + \dots + X_n) = V(X_1) + V(X_2) + \dots + V(X_n)$$

For any random variables (regardless of dependence!):

$$E(X_1 + X_2 + \dots + X_n) = E(X_1) + E(X_2) + \dots + E(X_n)$$

1.11 Special discrete distributions

The formulae for the PMF, expectation and variance of all the specific distributions below, apart from discrete uniform distribution, are given in the exam on the formulae sheet. It is therefore not needed to memorise them.

1.11.1 Discrete uniform distribution

A random variable X has a discrete uniform distribution if each of the n values in its range $x_1, x_2, \dots, x_n$ has equal probability.

Suppose the random variable X has uniform distribution. The PMF of X is given by ($X \sim \text{Uni}(n)$):

$$P(X = k) = f(k) = \frac{1}{n}$$

Suppose X is a discrete uniform random variable on the consecutive integers $a, a+1, a+2, \dots, b$ for $a \leq b$. Then:

$$\mu = \mathbb{E}[X] = \frac{a+b}{2}$$
$$\sigma^2 = \text{Var}(X) = \frac{(b-a+1)^2 - 1}{12}$$

1.11.2 Bernoulli distribution

In a random experiment, if each outcome consists of only two possibilities in each trial (such as in a toss of a coin), then such a random experiment is called a Bernoulli trial. Often one outcome is called ‘success’ and the other ‘failure’.

In a Bernoulli trial, let the possible events in each outcome be denoted by A and B , and let $P(A) = p$. Then:

- $A \cap B = \emptyset$
- $A \cup B = S$
- $P(B) = P(\bar{A}) = 1 - P(A) = 1 - p = q$

Let X be a random variable taking only two possible values $\{0, 1\}$. Let $p = P(X = 1)$. Then X is said to be a Bernoulli random variable with parameter p :

$$X \sim \text{Ber}(p)$$

$$E[X] = p$$

$$\text{Var}(X) = p(1 - p) = pq$$

1.11.3 Discrete binomial distribution

Consider a random experiment consisting of $n \in \mathbb{N}$ independent Bernoulli trials, each with probability of success p ($0 \leq p \leq 1$), constant across trials.

Let X denote the number of successes in these n trials. Then X is a Binomial random variable:

$$X \sim \text{Bin}(n, p)$$

$$P(X = k) = f(k) = \binom{n}{k} p^k (1 - p)^{n-k}, \quad k = 0, 1, \dots, n$$

$$\mathbb{E}[X] = np$$

$$\text{Var}(X) = np(1 - p)$$

1.11.4 Discrete geometric distribution

Consider a series of independent Bernoulli trials with constant probability of success p . Let the random variable X denote the number of trials until the first success.

Then X is called a geometric random variable with parameter p :

$$X \sim \text{Geom}(p)$$

$$P(X = k) = f(k) = p(1 - p)^{k-1}, \quad k = 1, 2, \dots$$

$$\mathbb{E}[X] = \frac{1}{p}$$

$$\text{Var}(X) = \frac{1 - p}{p^2}$$

The geometric distribution is the only discrete distribution satisfying the memoryless property. This means that the probability of success in the future does not depend on the past:

$$P(X > m + n \mid X > m) = P(X > n)$$

1.11.5 Negative binomial distribution

The negative binomial distribution generalises the geometric distribution.

Instead of counting the number of trials until the first success, count the number of trials until the r -th success occurs, where $r \in \mathbb{N}$.

Consider a sequence of independent Bernoulli trials, each with the same probability of success p . Let the random variable X denote the total number of trials needed to achieve r successes.

Then X is called a negative binomial random variable with parameters r and p :

$$\begin{aligned} X &\sim \text{NegBin}(r, p) \\ P(X = k) = f(k) &= \begin{cases} 0, & k < r \\ \binom{k-1}{r-1} p^r (1-p)^{k-r}, & k \geq r \end{cases}, \quad k \in \mathbb{N} \\ \mathbb{E}[X] &= \frac{r}{p} \\ \text{Var}(X) &= \frac{r(1-p)}{p^2} \end{aligned}$$

1.11.6 Poisson distribution

The Poisson distribution models the number of events occurring in a fixed interval of time or space. Instead of counting trials until success, it counts how many times an event occurs when events happen independently at a constant average rate $\lambda > 0$.

Consider a process where events occur randomly and independently, with an average rate of λ events per interval.

Let the random variable X denote the number of events in a given interval.

Then X is called a Poisson random variable with parameter λ :

$$\begin{aligned} X &\sim \text{Poisson}(\lambda) \\ P(X = k) = f(k) &= \begin{cases} \frac{\lambda^k e^{-\lambda}}{k!}, & k \geq 0 \\ 0, & k < 0 \end{cases}, \quad k \in \mathbb{N}_0 \\ \mathbb{E}[X] &= \lambda \\ \text{Var}(X) &= \lambda \end{aligned}$$

Key properties:

- Events occur independently
- The rate λ is constant over the interval
- The probability of more than one event in a very small interval is negligible
- The interarrival times of a Poisson process have a (continuous) exponential distribution with mean $\frac{1}{\lambda}$.

Take $Y \sim \text{Bin}(n, p)$ with large n , small p , such that $np \simeq \lambda$. Then Y is approximately $\text{Poisson}(\lambda)$ distributed

- this means the Poisson distribution can be used to approximate the binomial distribution

1.12 Continuous distributions

A random variable X is continuous if it can take infinitely many values within a range (including decimals).

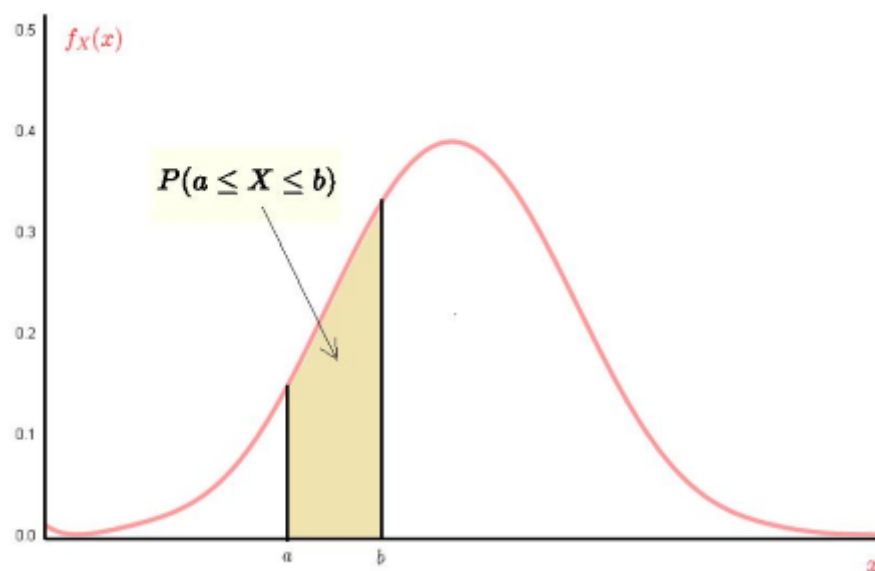
Since it can have infinitely many values, the probability of X taking any specific value is 0. Therefore, rather than considering the probability of X taking a particular value, we consider the probability of X lying within a certain interval.

1.12.1 Probability density function

A probability density function (PDF) describes how a continuous random variable is distributed.

The PDF of X is denoted by f_X or simply by f .

The probability that the random variable lies within an interval is given by the area under the curve of the PDF over that interval (and thus can be obtained using integration).



For a continuous random variable X , a probability density function is a function $f(x)$ such that:

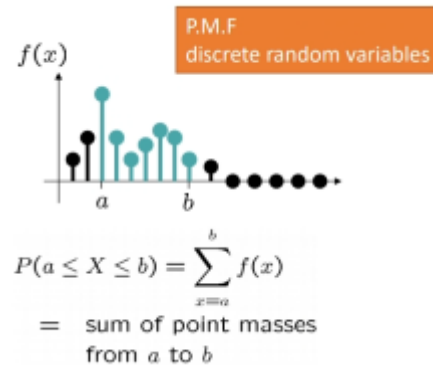
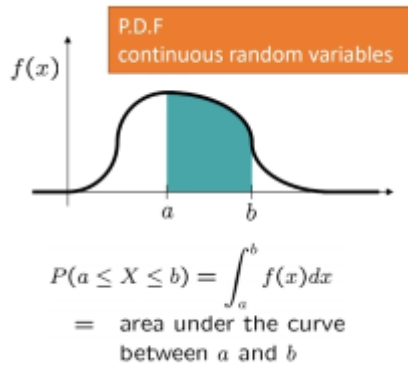
1. $f(x) \geq 0$, for all $x \in \mathbb{R}$
2. $\int_{-\infty}^{+\infty} f(x) dx = 1$
3. $P(a \leq X \leq b) = \int_a^b f(x) dx$, for any $a, b \in \mathbb{R}$ such that $a \leq b$

Note that since the probability of X taking any specific value is 0:

$$P(a \leq X \leq b) = P(a < X \leq b) = P(a < X < b) = P(a \leq X < b)$$

1.12.2 PDF and PMF

The PMF for discrete variables and the PDF for continuous variables play a similar role: they both describe how probabilities are distributed, but one uses sums and the other uses integrals.



1.12.3 Cumulative distribution function

Just as for discrete variables, the cumulative distribution function (CDF) for a continuous variable gives the probability that a random variable takes a value less than or equal to a certain number.

CDF is a function $F : \mathbb{R} \rightarrow [0, 1]$ that gives the probability that X is less than or equal to a given value x . It is defined as follows:

$$F(X) = P(X \leq x) = \int_{-\infty}^x f(t)dt$$

As in the discrete case, a continuous CDF has the following properties:

- Non-decreasing: if $x \leq y$, then $F(x) \leq F(y)$
- $0 \leq F(x) \leq 1$, $\lim_{x \rightarrow -\infty} F(x) = 0$ and $\lim_{x \rightarrow +\infty} F(x) = 1$
- for $a \leq b$, $P(a \leq X \leq b) = F(b) - F(a)$

For a continuous variable, the derivative of the CDF is the PDF.

1.12.4 Mean and variance

As with discrete variables, mean and variance are used to summarise the distribution:

$$\mu = E(X) = \int_{-\infty}^{+\infty} xf(x)dx$$

$$\sigma^2 = E[(X - \mu)^2] = \int_{-\infty}^{+\infty} (x - \mu)^2 f(x) dx \quad , \text{ or equivalently}$$

$$\sigma^2 = E[X^2] - \mu^2 = \int_{-\infty}^{+\infty} x^2 f(x)dx - \mu^2$$

1.12.5 Functions of random continuous variables

Let X be a random variable and let $h : \mathbb{R} \rightarrow \mathbb{R}$ be an arbitrary function, then $Y = h(X)$ is also a random variable.

The properties are the same for continuous variables as for **discrete variables**, but the mean is calculated using an integral rather than a sum:

$$E[Y] = E[h(X)] = \int_{-\infty}^{+\infty} h(x)f(x) dx$$

These properties include:

- $E(aX + b) = aE(X) + b$
- $V(aX + b) = a^2V(X)$
- $V(X) = E[X^2] - (E[X])^2$

1.12.6 Specific continuous distributions

The formulae for the PDF, expectation and variance of all the specific distributions below, are given in the exam on the formulae sheet. It is therefore not needed to memorise them.

Continuous uniform distribution A continuous uniform distribution models a random variable that is equally likely to take any value within a given interval $[a, b]$.

Let X be a continuous random variable with density function:

$$f(x) = \begin{cases} \frac{1}{b-a}, & a \leq x \leq b \\ 0, & \text{otherwise} \end{cases}$$

Then, X is said to be a continuous uniform random variable with range $[a, b]$:

$$\begin{aligned} \mathbb{E}[X] &= \frac{a+b}{2} \\ \mathbb{E}[X^2] &= \frac{a^2 + ab + b^2}{3} \\ \text{Var}(X) &= \frac{(b-a)^2}{12} \end{aligned}$$

Exponential distribution A continuous random variable X is called an exponential random variable with parameter $\lambda > 0$, (denoted by $X \sim \text{Exp}(\lambda)$) when its probability density function is:

$$f(x) = \begin{cases} \lambda e^{-\lambda x}, & x \geq 0 \\ 0, & x < 0 \end{cases}$$

Here, λ is called the rate parameter, and is the average number of events occurring in one period of unit time. This λ is the same as in the **Poisson distribution**; a variable $X \sim \text{Exp}(\lambda)$ models the time between events in a Poisson process with rate λ .

For the exponential distribution:

$$\begin{aligned} \mathbb{E}[X] &= \frac{1}{\lambda} \\ \mathbb{E}[X^2] &= \frac{2}{\lambda^2} \end{aligned}$$

$$\text{Var}(X) = \frac{1}{\lambda^2}$$

The exponential distribution is the only continuous distribution that satisfies the memoryless property:

- For any $t_1, t_2 \geq 0$:

$$P(X > t_1 + t_2 \mid X > t_1) = P(X > t_2)$$

- If X denotes the time until a certain event happens, if we have already waited t_1 for the event, then the probability that we must wait t_2 more time does not depend on the value of t_1

1.13 The normal distribution

The normal distribution (also called the Gaussian distribution) is a continuous distribution with significant importance, describing data that clusters around a central value with a symmetric ‘bell-shaped’ curve.

The sum (or average) of many independent random variables tends to look normal (like a bell curve), regardless of the original distributions.

A random variable X is said to be normally distributed if:

$$X \sim \mathcal{N}(\mu, \sigma^2)$$

where:

- $\mu \in (-\infty, \infty)$ is the mean (centre)
- $\sigma^2 > 0$ is the variance (spread)

and its probability density function (PDF) is:

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

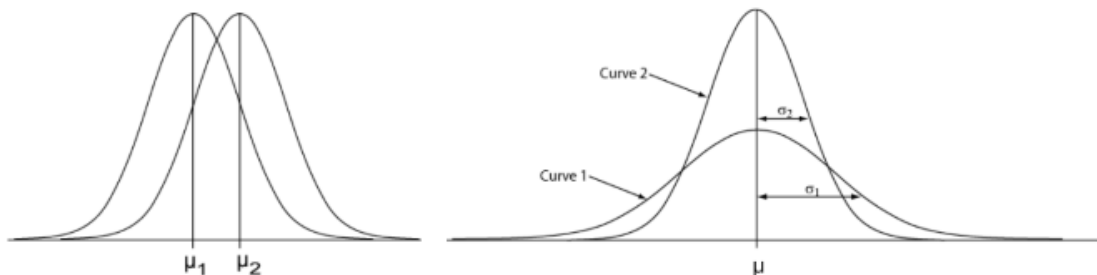
For a normally distributed variable X :

$$E(X) = \mu$$

$$\text{Var}(X) = \sigma^2$$

It forms a bell shaped curve which is symmetric about μ :

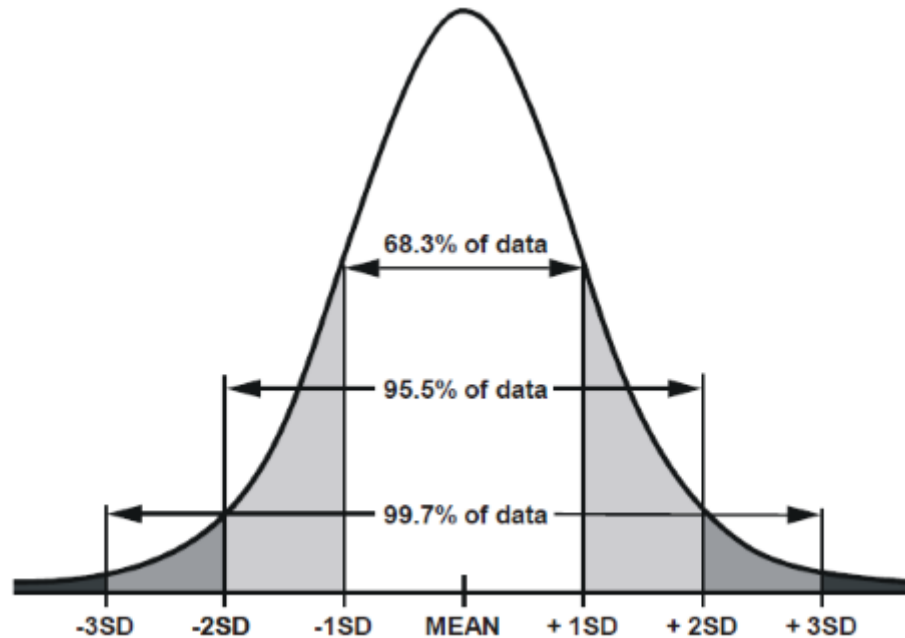
- μ controls the position of the curve (shifts it left/right)
- σ controls the spread of the curve (larger $\sigma \rightarrow$ wider and flatter curve)
- the total area under the curve is 1



1.13.1 The empirical rule

For a normal distribution: - About 68% of values (area) lie within $\mu \pm \sigma$ - About 95% lie within $\mu \pm 2\sigma$ - About 99.7% lie within $\mu \pm 3\sigma$

These results are known as the empirical rule.



1.13.2 Standard normal distribution

The best way to find a probability in a normal distribution is to convert it to the standard normal distribution.

A random variable Z is said to have a standard normal distribution if:

$$Z \sim \mathcal{N}(0, 1)$$

where:

- the mean (μ) is 0
- the variance (σ^2) is 1

and its probability density function (PDF) is:

$$\phi(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}}$$

For a standard normally distributed variable Z :

$$E(Z) = 0$$

$$\text{Var}(Z) = 1$$

All normal distributions X can be converted into the standard normal distribution Z by subtracting the mean and dividing by the standard deviation:

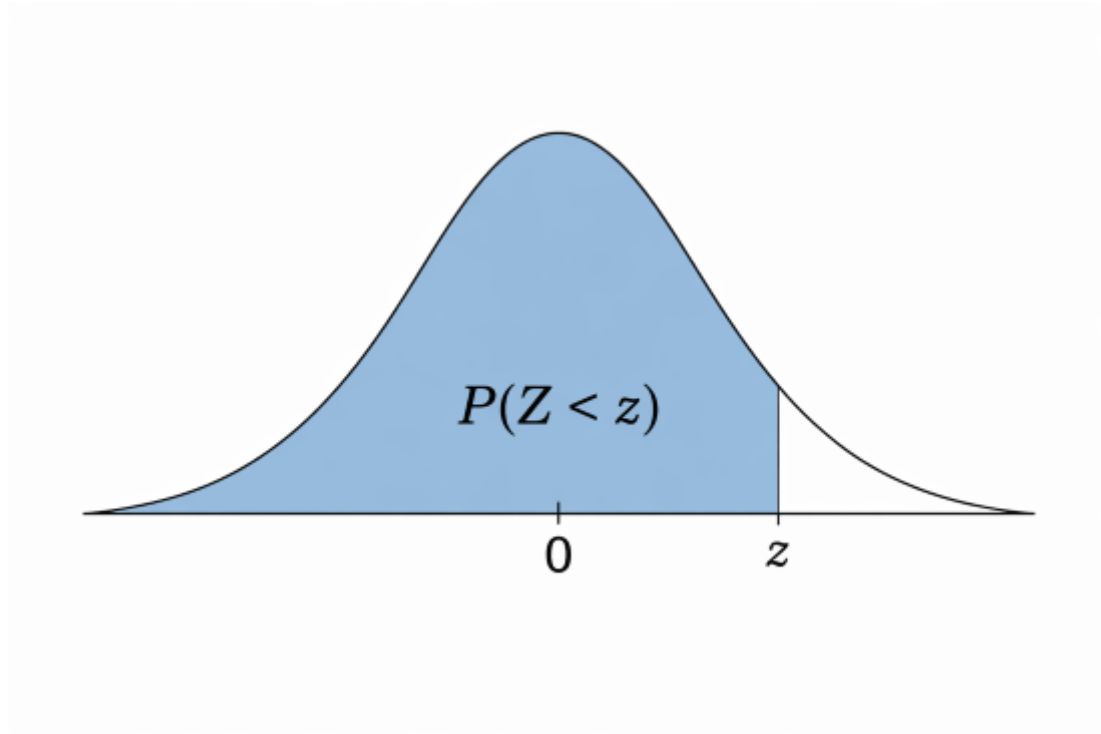
$$Z = \frac{X - \mu}{\sigma}$$

- this is called Z-score:
 - it gives the number of standard deviations the value of X is above or below the mean

The standard normal distribution makes it much easier to compare values from normal distributions using different scales, since they can be converted to use the same scale.

Z-table Integrals for the standard normal distribution have been computed and placed in a table called the Z-table, meaning that it is not necessary to compute integrals in order to determine probabilities after converting to the standard normal distribution; the probabilities can be looked up in the table.

The Z-table provided in the exam gives the area to the to the left of each Z-score.



To read the Z-table:

- Split the z-score after the first decimal point
 - the first digit and first decimal give the row
 - the second decimal gives the column
- Identify the value in this row and column
- This value gives the total area to the left of the z-score
 - this is $P(Z < z)$
 - $P(Z > z) = 1 - P(Z < z)$
 - $P(a < Z < b) = P(Z < b) - P(Z < a)$

STANDARD NORMAL DISTRIBUTION: Table Values Represent AREA to the LEFT of the Z score.

Z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
-3.9	.00005	.00005	.00004	.00004	.00004	.00004	.00004	.00004	.00003	.00003
-3.8	.00007	.00007	.00007	.00006	.00006	.00006	.00006	.00005	.00005	.00005
-3.7	.00011	.00010	.00010	.00010	.00009	.00009	.00008	.00008	.00008	.00008
-3.6	.00016	.00015	.00015	.00014	.00014	.00013	.00013	.00012	.00012	.00011
-3.5	.00023	.00022	.00022	.00021	.00020	.00019	.00019	.00018	.00017	.00017
-3.4	.00034	.00032	.00031	.00030	.00029	.00028	.00027	.00026	.00025	.00024
-3.3	.00048	.00047	.00045	.00043	.00042	.00040	.00039	.00038	.00036	.00035
-3.2	.00069	.00066	.00064	.00062	.00060	.00058	.00056	.00054	.00052	.00050
-3.1	.00097	.00094	.00090	.00087	.00084	.00082	.00079	.00076	.00074	.00071
-3.0	.00135	.00131	.00126	.00122	.00118	.00114	.00111	.00107	.00104	.00100
-2.9	.00187	.00181	.00175	.00169	.00164	.00159	.00154	.00149	.00144	.00139
-2.8	.00256	.00248	.00240	.00233	.00226	.00219	.00212	.00205	.00199	.00193
-2.7	.00347	.00336	.00326	.00317	.00307	.00298	.00289	.00280	.00272	.00264
-2.6	.00466	.00453	.00440	.00427	.00415	.00402	.00391	.00379	.00368	.00357
-2.5	.00621	.00604	.00587	.00570	.00554	.00539	.00523	.00508	.00494	.00480
-2.4	.00820	.00798	.00776	.00755	.00734	.00714	.00695	.00676	.00657	.00639
-2.3	.01072	.01044	.01017	.00990	.00964	.00939	.00914	.00889	.00866	.00842
-2.2	.01390	.01355	.01321	.01287	.01255	.01222	.01191	.01160	.01130	.01101
-2.1	.01786	.01743	.01700	.01659	.01618	.01578	.01539	.01500	.01463	.01426
-2.0	.02275	.02222	.02169	.02118	.02068	.02018	.01970	.01923	.01876	.01831

The image shows an extract from the Z-table. The boxed value corresponds to $P(Z < -2.32)$.

1.13.3 Normal approximation to the binomial distribution

If $X \sim \text{Bin}(n, p)$ then for large n , the distribution of X looks approximately normal:

$$Y \sim \mathcal{N}(np, np(1-p)) \quad \text{with } X \approx Y$$

This approximation works well when:

- $np > 5$
- $n(1-p) > 5$

This ensures the binomial distribution is not too skewed.

Since the binomial distribution is discrete, while the normal distribution is continuous, a continuity correction should be applied when using normal approximation:

$$P(X \leq k) \approx P(Y \leq k + 0.5)$$

$$P(X < k) \approx P(Y \leq k - 0.5)$$

1.13.4 Normal approximation to Poisson distribution

If $X \sim \text{Poisson}(\lambda)$ then for large λ , the distribution of X looks approximately normal:

$$Y \sim \mathcal{N}(\lambda, \lambda) \quad \text{with } X \approx Y$$

A continuity correction must also be applied in the same way as with approximating a binomial distribution.

1.14 Multiple random variables

Multiple random variables are used to model and calculate the probability of events that depend on or occur together, capturing relationships between them rather than treating them in isolation.

Joint distributions enable reasoning about the relationship between multiple variables:

- whether or not they are dependent or independent
- conditional probabilities

Distributions with two variables are called bivariate distributions. Distributions with more than two variables are called multivariate distributions.

1.14.1 Discrete bivariate joint distribution

The probability mass function (PMF) of a single discrete random variable X specifies how much probability mass is placed on each possible X value.

- The joint PMF of two discrete random variables X and Y describes how much probability mass is placed on each possible pair of values (x, y)

If X and Y are discrete random variables, the function given by

$$f(x, y) = P(X = x, Y = y)$$

for each pair of values (x, y) is called the joint probability distribution (joint probability mass function) of X and Y . It has the following properties:

- $f(x, y) \geq 0$
- $\sum_x \sum_y f(x, y) = 1$
- $f(x, y) = P(X = x, Y = y) \equiv P(\{X = x\} \cap \{Y = y\})$

The joint CDF is defined by:

$$F(x, y) = P(X \leq x, Y \leq y) = \sum_{s \leq x} \sum_{t \leq y} f(s, t), \quad -\infty < x, y < \infty$$

1.14.2 Discrete multivariate joint distributions

The concept of a joint distribution extends to more than two random variables.

- If $X_1, X_2, \dots, X_n$ are all discrete random variables, the joint PMF is:

$$f(x_1, x_2, \dots, x_n) = P(X_1 = x_1, X_2 = x_2, \dots, X_n = x_n)$$

1.14.3 Discrete marginal probability distributions

A marginal probability distribution is the distribution of a single random variable obtained from a joint distribution by removing the other variable(s).

For a discrete bivariate joint distribution, this is done by summation:

- Let X and Y be two discrete random variables with joint probability mass function $f_{XY}(x, y)$. Then, the corresponding marginal probability mass functions are given by:

$$f_X(x) = P(X = x) = \sum_y P(X = x, Y = y) = \sum_y f_{XY}(x, y)$$

$$f_Y(y) = P(Y = y) = \sum_x P(X = x, Y = y) = \sum_x f_{XY}(x, y)$$

1.14.4 Discrete conditional distributions

A conditional distribution describes the probability distribution of one random variable given that another random variable has taken a specific value.

Let X and Y be two discrete random variables with joint probability mass function $f_{XY}(x, y)$.

Assume $f_X(x) > 0$ (meaning $X = x$ is possible), then:

$$f_{Y|X=x}(y) = P(Y = y | X = x) = \frac{P(X = x, Y = y)}{P(X = x)} = \frac{f_{XY}(x, y)}{f_X(x)}$$

Similarly, if $f_Y(y) > 0$:

$$f_{X|Y=y}(x) = P(X = x | Y = y) = \frac{P(X = x, Y = y)}{P(Y = y)} = \frac{f_{XY}(x, y)}{f_Y(y)}$$

The conditional distribution is a valid PMF, so: 1. $\sum_{y \in Y} f_{Y|X=x}(y) = 1$

2. For any set B :

$$P(Y \in B | X = x) = \sum_{y \in B} f_{Y|X=x}(y)$$

1.14.5 Continuous bivariate joint distributions

The probability density function (PDF) of a single continuous random variable X describes how probability density is distributed over possible values of X .

- The joint PDF of two continuous random variables X and Y describes how density is distributed over pairs (x, y)

If X and Y are continuous random variables, the function

$$f_{XY}(x, y)$$

is called the joint probability density function of X and Y , where probabilities are obtained by integration:

$$P(a \leq X \leq b, c \leq Y \leq d) = \int_a^b \int_c^d f_{XY}(x, y) dy dx$$

It has the following properties: - $f_{XY}(x, y) \geq 0$ - $\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_{XY}(x, y) dx dy = 1$ - $P((X, Y) \in R) = \iint_R f_{XY}(x, y) dy dx$

1.14.6 Continuous multivariate joint distributions

- If $X_1, X_2, \dots, X_n$ are all continuous random variables, the joint PDF is the function $f(x_1, x_2, \dots, x_n)$ such that for any n intervals $[a_1, b_1], \dots, [a_n, b_n]$:

$$P(a_1 \leq X_1 \leq b_1, \dots, a_n \leq X_n \leq b_n) = \int_{a_1}^{b_1} \dots \int_{a_n}^{b_n} f(x_1, x_2, \dots, x_n) dx_n \dots dx_1$$

1.14.7 Continuous marginal probability distributions

For a continuous joint distribution, a distribution of a single random variable can be obtained by integrating over the other variable(s):

- Let X and Y be two continuous random variables with joint probability density function $f_{XY}(x, y)$. Then, the corresponding marginal PDFs are given by:

$$f_X(x) = \int f_{XY}(x, y) dy$$

$$f_Y(y) = \int f_{XY}(x, y) dx$$

1.14.8 Continuous conditional distributions

Let X and Y be two continuous random variables with joint probability density function $f_{XY}(x, y)$.

Assume $f_X(x) > 0$ (meaning $X = x$ is possible), then:

$$f_{Y|X=x}(y) = \frac{f_{XY}(x, y)}{f_X(x)}$$

Similarly, if $f_Y(y) > 0$:

$$f_{X|Y=y}(x) = \frac{f_{XY}(x, y)}{f_Y(y)}$$

The conditional distribution is a valid PDF, so: 1. $\int_{-\infty}^{\infty} f_{Y|X=x}(y) dy = 1$

2. For any set $B \subseteq \mathbb{R}$:

$$P(Y \in B | X = x) = \int_B f_{Y|X=x}(y) dy$$

1.14.9 Independence and dependence

Let X and Y be two random variables (discrete or continuous) with possible values $\mathcal{X}$ and $\mathcal{Y}$ respectively.

They are independent if the following equivalent statements hold:

- $f_{XY}(x, y) = f_X(x) f_Y(y)$ for all $x \in \mathcal{X}, y \in \mathcal{Y}$
- $f_{Y|X=x} = f_Y(y)$ for all x, y with $f_X(x) > 0$
- $f_{X|Y=y} = f_X(x)$ for all x, y with $f_Y(y) > 0$
- $P(X \in A, Y \in B) = P(X \in A) P(Y \in B)$ for all sets A, B

1.14.10 Mean and variance of joint random variables

To find the mean or variance of a single random variable in a joint distribution, first obtain its marginal distribution, then compute the expectation or variance as in the standard, single-variable case.

Properties of mean and variance For any random variables X and Y and constants a , b , and c : - $E[a] = a$

- $E[X + Y] = \mu_X + \mu_Y$
- $E[aX + bY + c] = a\mu_X + b\mu_Y + c$
- $E[h(X, Y)] = \sum_x \sum_y h(x, y) f_{XY}(x, y)$ (discrete)
- $E[h(X, Y)] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} h(x, y) f_{XY}(x, y) dx dy$ (continuous)
- $V(X) = E[X^2] - \mu_X^2$
- $V(X + Y) = \sigma_X^2 + \sigma_Y^2 + 2\sigma_{XY}$
- $V(aX + bY + c) = a^2\sigma_X^2 + b^2\sigma_Y^2 + 2ab\sigma_{XY}$

1.14.11 Covariance

Covariance measures the direction and strength of the linear relationship between two variables.

The covariance between two discrete or continuous random variables X and Y is denoted by $\text{Cov}(X, Y)$ and is given by:

$$\text{Cov}(X, Y) = \sigma_{XY} = E[XY] - E[X]E[Y]$$

Alternatively,

$$\text{Cov}(X, Y) = E[(X - \mu_X)(Y - \mu_Y)]$$

where $\mu_X = E[X]$ and $\mu_Y = E[Y]$.

Properties of covariance For any random variables X and Y and constants a , b , and c :

- $\text{Cov}(aX + b, cY + d) = ac \text{Cov}(X, Y)$
- $\text{Cov}(X_1 + X_2, Y) = \text{Cov}(X_1, Y) + \text{Cov}(X_2, Y)$
- $\text{Cov}(X, X) = \text{Var}(X)$

If X and Y are independent then $\text{Cov}(X, Y) = 0$

- The converse is not necessarily true - if covariance is 0 the variables might not be independent

1.15 Correlation coefficient

The correlation coefficient also measures the strength and direction of the linear relationship between two random variables, but in a standardised way. It is the normalised covariance.

The correlation coefficient between two random variables X and Y is:

$$\text{Corr}(X, Y) = \rho_{XY} = \frac{\text{Cov}(X, Y)}{\sqrt{V(X)V(Y)}}$$

It satisfies:

$$-1 \leq \rho_{XY} \leq 1$$

- $\rho = 1$ if and only if $Y = aX + b$ with $a > 0$
- $\rho = -1$ if and only if $Y = aX + b$ with $a < 0$

If X and Y are independent random variables then they are uncorrelated, i.e., $\rho_{XY} = 0$

- the converse is not necessarily true - zero correlation need not imply independence

1.16 Population and sample

A **population** is the complete set of all individuals, objects, or groups that the researcher wants to study.

A **sample** is a subset of the population, ideally selected so that it is representative of the population (often through random sampling).

A numerical characteristic of a population is called a parameter. For example:

- μ : population mean
- σ : population standard deviation
- p : population proportion (the fraction of the population with a given characteristic)

A numerical characteristic computed from a sample is called a statistic. For example:

- $\bar{x}$: sample mean
- s : sample standard deviation
- $\hat{p}$: sample proportion

A statistic is a random variable because its value depends on the random sample selected:

- a sample contains only part of the population
- the observed value of a statistic depends on the particular sample that happens to be selected
- there will be some variability in the value of a statistic over different occasions of sampling

1.16.1 Calculating mean and standard deviation

For a population of size N :

$$\mu = \frac{1}{N} \sum_{i=1}^N x_i$$

$$\sigma^2 = \frac{1}{N} \sum_{i=1}^N (x_i - \mu)^2$$

$$\sigma = \sqrt{\sigma^2}$$

For a sample of size n :

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$$

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$$

$$s = \sqrt{s^2}$$

1.16.2 Other useful statistics

There are several other useful statistics for analysing a sample of size n :

- median
 - the value that splits the ordered dataset into two halves, so that approximately half the observations are below it and half are above it
 - if n is odd, the median is the $\frac{n+1}{2}$ th ordered value
 - if n is even, the median is the average of the $\frac{n}{2}$ th and $(\frac{n}{2} + 1)$ th ordered values
- mode
 - the most frequently occurring value
 - there can be one mode, multiple modes, or no mode
- range
 - the difference between the largest value and the smallest value
- order statistics
 - the values of the sample after sorting them from smallest to largest
 - the k th order statistic ($X_{(k)}$) is the k th smallest observation

1.16.3 Population and sample distribution

The population distribution of a variable describes how the values of that variable are distributed across all individuals in the population.

The sampling distribution of a statistic is the probability distribution of the values taken by that statistic over all possible random samples of the same size n drawn from the population:

- each possible sample produces one value of the statistic
- the sampling distribution describes how those statistic values vary from sample to sample
- it is usually derived theoretically by considering all possible samples, or approximated in practice by repeatedly drawing many random samples
- sampling is often modeled as being done with replacement (after each observation is selected, it is returned to the population before the next selection)

1.16.4 Statistical inference

The process of statistical inference involves using information from a sample to draw conclusions about a wider population:

- data is collected from a representative sample of the population
- the sample data is used to estimate population parameters and assess how reliable those estimates are
- conclusions about the population are made based on the observed sample

1.16.5 Markov's inequality

Let X be a non-negative random variable:

$$P(X \geq a) \leq \frac{E(x)}{a}, \forall a > 0$$

Markov's inequality gives an upper bound on the probability that a non-negative random variable is at least as large as some positive value a . If the expected value $E(X)$ is small compared with a , then the probability that X exceeds a must also be small:

- the probability that X is bigger than a cannot be more than its average value divided by a

1.16.6 Chebyshev's inequality

Let X be a random variable with $E(X) = \mu$ and $V(X) = \sigma^2$:

$$P(|X - \mu| \geq k) \leq \frac{\sigma^2}{k^2}, \forall k > 0$$

Chebyshev's inequality describes how concentrated a random variable is around its mean using only its mean and variance.

- the probability that X deviates far from its mean is bounded above by a quantity that depends only on the variance.
- larger variance means a weaker (looser) bound; smaller variance means stronger concentration around the mean.

1.16.7 Sampling distribution

Suppose that $\bar{X}$ is the mean of a sample of size n drawn from a large population with mean μ and standard deviation σ .

Then the sampling distribution of $\bar{X}$ has mean μ and standard deviation $\frac{\sigma}{\sqrt{n}}$.

$$E(\bar{X}) = E\left(\frac{X_1 + X_2 + \dots + X_n}{n}\right) = \mu$$

$$V(\bar{X}) = V\left(\frac{X_1 + X_2 + \dots + X_n}{n}\right) = \frac{\sigma^2}{n}$$

$$\text{std}(\bar{X}) = \frac{\sigma}{\sqrt{n}}$$

This means that:

- if we repeatedly take random samples of size n , the average of all sample means equals the true population mean μ
- the sample mean becomes less variable as n increases, because averaging independent observations reduces random noise

When sampling is done from a normal distribution, then $\bar{X}$ also has a normal distribution (for each value of n).

1.16.8 Weak law of large numbers

Let $X_1, X_2, \dots, X_n$ be independent and identically distributed random variables with $E(X_i) = \mu$ and $V(X_i) = \sigma^2$.

Let the sample mean be:

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i.$$

Then for any $\varepsilon > 0$,

$$P(|\bar{X}_n - \mu| \geq \varepsilon) \rightarrow 0 \quad \text{as } n \rightarrow \infty.$$

This means that:

- the sample mean $\bar{X}_n$ becomes increasingly close to the population mean μ as the sample size increases
- the probability of large deviations goes to zero

1.16.9 Strong law of large numbers

Let $X_1, X_2, \dots, X_n$ be independent and identically distributed random variables with $E(X_i) = \mu$.

Let the sample mean be:

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i.$$

Then:

$$\bar{X}_n \xrightarrow{a.s.} \mu$$

Equivalently,

$$P\left(\lim_{n \rightarrow \infty} \bar{X}_n = \mu\right) = 1.$$

This means that:

- for almost every infinite sequence of samples, the sample mean converges to μ
- the running average stabilises at the true mean with probability 1

1.16.10 Central limit theorem

Consider independent and identically distributed random variables $X_1, X_2, \dots, X_n$ all distributed with $E(X_i) = \mu$ and $V(X_i) = \sigma^2 < \infty$.

Let the sample mean be:

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i.$$

Then, as $n \rightarrow \infty$,

$$\bar{X}_n \approx N\left(\mu, \frac{\sigma^2}{n}\right)$$

More precisely:

$$\frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \xrightarrow{d} N(0, 1).$$

This means that:

- for large n , the distribution of the sample mean is approximately normal, regardless of the original distribution of X_i
- the mean of the sampling distribution is μ
- the variance of the sampling distribution is $\frac{\sigma^2}{n}$
- the sample mean can be standardised to follow an approximate standard normal distribution

In general:

- there is no fixed sample size that guarantees normality, however:
 - a sample size of 25 is usually enough to obtain a normal sampling distribution from a strongly skewed distribution with mild outliers
 - a sample size of 40 is usually good enough to overcome extreme skewness and extreme outliers

Importance of the central limit theorem The central limit theorem (CLT) explains why the normal distribution appears so frequently in the real world:

- many real-world quantities are formed from the sum or average of many small, independent random effects

The CLT also explains why certain discrete distributions can be approximated by the normal distribution when parameters are large, such as:

- the Binomial distribution (when n is large and p is not too close to 0 or 1)
- the Poisson distribution (when λ is large)

In these cases, sums of many small random contributions produce an approximately normal distribution.

1.16.11 Standard error

The standard error of a statistic is the standard deviation of its sampling distribution.

For the sample mean:

$$\sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}}$$

For the sample proportion:

$$\sigma_{\hat{p}} = \sqrt{\frac{p(1-p)}{n}}$$

1.16.12 Types of statistical inference

There are two main types of statistical inference:

- **Estimation:** using sample data to estimate unknown population parameters
 - point estimation: using a sample statistic to estimate a population parameter
 - interval estimation: calculating a range of plausible values for a population parameter (a confidence interval)
- **Hypothesis testing:** using sample data to assess whether there is enough evidence to support or reject a claim about a population parameter

1.16.13 Bias and variance of an estimator

Let $\hat{\theta}$ be an estimator of a population parameter θ .

The bias of the estimator is defined as:

$$b(\hat{\theta}) = E(\hat{\theta}) - \theta$$

- if $b(\hat{\theta}) = 0$, the estimator is said to be unbiased

The variance of an estimator measures how much the estimator varies from sample to sample:

$$V(\hat{\theta}) = E \left[(\hat{\theta} - E(\hat{\theta}))^2 \right]$$

- a smaller variance means the estimator is more stable across repeated samples

A good estimator has both low bias and low variance.

1.16.14 Confidence intervals

A confidence interval gives a range of plausible values for an unknown population parameter.

Each confidence interval is associated with a confidence level, which describes the long-run performance of the method used to construct the interval:

- for example, a 95% confidence interval means that if we repeatedly took samples and constructed a confidence interval from each sample, then approximately 95% of those intervals would contain the true population parameter.

A confidence interval is generally given by:

$$\text{Confidence interval} = \text{Point estimate} \pm \text{Critical value} \times \text{Standard error}$$

where:

- the point estimate is the sample statistic used to estimate the parameter
- the critical value is determined by the chosen confidence level (e.g. 1.96 for a 95% confidence interval using the standard normal distribution)
- the standard error measures the variability of the estimator

Determining critical value The critical value used is determined by the chosen confidence level.

Suppose we want the critical value for a 95% confidence level.

This means that the true parameter lies within the confidence interval 95% of the time. Therefore, using the standard normal distribution, we want:

$$P(|Z| < z) = 0.95$$

The amount of area in each tail is

$$\frac{1 - 0.95}{2} = \frac{0.05}{2} = 0.025$$

Therefore, we need z such that:

$$P(Z < z) = 0.95 + 0.025 = 0.975$$

Looking this up in the z -table gives $z = 1.96$. Therefore, 1.96 is the critical value.

One-sided confidence intervals In some cases, we are only interested in an upper or lower bound for the parameter, rather than an interval.

A one-sided confidence interval has the form:

- Upper bound: $(-\infty, \text{upper limit})$
- Lower bound: $(\text{lower limit}, \infty)$

For a 95% one-sided confidence interval, the entire 5% significance level is placed in only one tail.

Therefore:

- Upper bound case:

$$P(Z < z) = 0.95$$

- Lower bound case:

$$P(Z > z) = 0.95 \quad \text{or equivalently} \quad P(Z < z) = 0.05$$

From the standard normal table:

- 95th percentile: $z \approx 1.645$

So the critical value for a 95% one-sided confidence interval is:

$$z = 1.645$$

1.17 Hypothesis testing

Hypothesis testing is a statistical method that uses sample data to evaluate a hypothesis about a population parameter.

It involves two parts:

- a hypothesis
 - a hypothesis is an assumption about a population parameter
- a decision process to determine whether there is enough evidence to reject the hypothesis

It is used to help differentiate between real effects and random variation in data.

1.17.1 Hypotheses

In hypothesis testing there are two hypotheses

- **Null hypothesis, H_0**
 - the statement being tested
 - assumes no effect, no difference, or no change
 - typically states that a population parameter equals a specific value
 - e.g. $\mu = 100$
- **Alternative hypothesis, H_1**
 - the statement that contradicts the null hypothesis
 - the claim for which we seek evidence
 - represents the presence of an effect or difference
 - e.g. $\mu \neq 100$

1.17.2 Hypothesis tests

A hypothesis test has the following general outline:

1. State the null hypothesis H_0 and alternative hypothesis H_1 , both about a population parameter θ .
2. Take a random sample and compute the sample statistic $\hat{\theta}$.
3. Assuming H_0 is true, calculate the probability of obtaining a result as extreme as, or more extreme than, the observed value of $\hat{\theta}$. This probability is called the p-value.
4. If the p-value is smaller than the chosen significance level, reject H_0 .
5. Otherwise, there is insufficient evidence to reject H_0 .

1.17.3 Test statistic and rejection criteria

For each hypothesis test, the following must be specified:

- Test statistic
 - a value computed from the sample data and used to assess the null hypothesis
 - measures how far the observed sample statistic is from the value predicted by H_0
 - typically obtained by standardising the sample statistic under the assumption that H_0 is true
 - common examples include:
 - * z-score when the standard normal distribution is used
 - * t-score when Student's t -distribution is used
- Rejection criterion for H_0
 - the rule that determines when the null hypothesis should be rejected
 - based on the chosen significance level α
 - * the significance level is the probability that the test statistic will fall in the critical region when H_0 is actually true
 - may be expressed in either of two equivalent ways:
 - * comparing the p-value to α
 - * checking whether the test statistic falls in the critical region
 - the critical region is the set of values of the test statistic that lead to rejection of H_0
 - it can be two-tailed, upper-tailed or lower-tailed
 - a critical value is a value that separates the rejection region from the non-rejection region
 - * obtained from the appropriate distribution table (e.g. Z-table or t-table)

Test statistics for Z-test The Z-test (standard normal distribution) is the most commonly used test for hypothesis testing.

When testing for a population proportion, the test statistic is:

$$z = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}}$$

where:

- $\hat{p}$ is the sample proportion
- p_0 is the population proportion specified in the null hypothesis
- n is the sample size

When testing a hypothesis about a population mean, the test statistic is:

$$z = \frac{\bar{x} - \mu_0}{\frac{\sigma}{\sqrt{n}}}$$

where:

- $\bar{x}$ is the sample mean
- μ_0 is the population mean specified in the null hypothesis
- σ is the population standard deviation (assumed known)

- n is the sample size

Both follow the same structure:

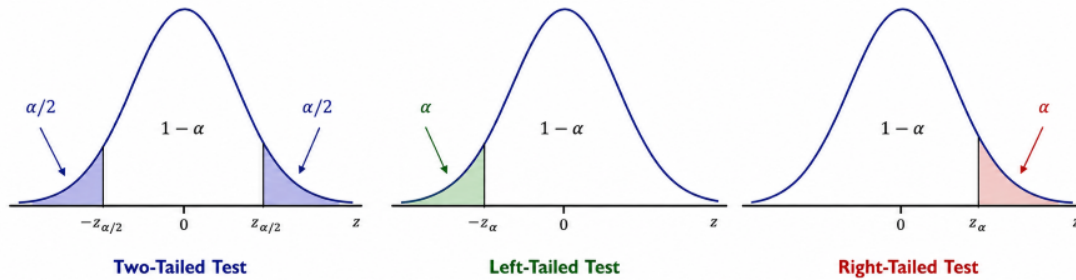
$$Z = \frac{\text{estimate} - \text{null value}}{\text{standard error}}$$

1.17.4 Critical region location

The tails of a distribution are the extreme regions beyond the critical value(s). If the calculated test statistic falls within one of these regions, the null hypothesis is rejected.

The number and location of the tails depend on the alternative hypothesis:

- $H_0 : \mu = \mu_0, \quad H_1 : \mu \neq \mu_0$
 - two-tailed test
 - the significance level α is split equally between the two tails
- $H_0 : \mu = \mu_0, \quad H_1 : \mu < \mu_0$
 - left-tailed test
 - the entire significance level α is placed in the left tail
- $H_0 : \mu = \mu_0, \quad H_1 : \mu > \mu_0$
 - right-tailed test
 - the entire significance level α is placed in the right tail



1.17.5 p-value

An alternative method to identifying the critical region is to calculate the p-value (probability value).

The p-value is the probability, assuming that H_0 is true, of obtaining a test statistic at least as extreme as the one observed from the sample data.

- if $p \leq \alpha$, reject H_0
- else, there is insufficient evidence to reject H_0

Calculating p-value To calculate the p-value:

- calculate the test statistic
- identify whether the test is two-tailed or one-tailed
 - **Right-tailed test** ($H_1 : \theta > \theta_0$)
 - * p-value is the area to the right of the test statistic
 - **Left-tailed test** ($H_1 : \theta < \theta_0$)

- * p-value is the area to the left of the test statistic
- **Two-tailed test** ($H_1 : \theta \neq \theta_0$)
 - * if the test statistic is positive:
 - p-value = twice the area to the right of the statistic
 - * if the test statistic is negative:
 - p-value = twice the area to the left of the statistic

1.17.6 Hypothesis testing with unknown standard deviation

When calculating the test statistic for the Z-test, the population standard deviation σ is used to calculate the standard error.

However, sometimes σ is not known. In this case:

- if $n > 30$, the z-distribution can still be used
- this is because with a larger sample size, the distribution of the sample is approximately normal (by the CLT)
 - the standard deviation of the sample s is used as the population standard deviation

$$z = \frac{\bar{X} - \mu_{\bar{X}}}{\frac{s}{\sqrt{n}}}$$

- if $n < 30$, the Student's t-distribution should be used instead
 - the test statistic is calculated in the same way, but the distribution is different

$$t = \frac{\bar{X} - \mu_{\bar{X}}}{\frac{s}{\sqrt{n}}}$$

Student's t-distribution The Student's t-distribution is a distribution that can be used when the population standard deviation is unknown and the number of samples is small:

- it corrects for the fact that the sample distribution is not normal
- bell shaped like the normal distribution, but with a lower peak and wider tails than the normal distribution, especially for very small samples
 - this accounts for the fact that small samples will produce extremely inaccurate estimates more often
- the distribution is different for different sample sizes
- has a mean of $t = 0$
- standard deviation varies with the sample size and is greater than 1 (unlike standard normal, where it is always 1)
- as the sample size n gets larger, the Student t-distribution gets closer to the standard normal distribution
 - this is why, for $n > 30$, the z-distribution can be used

1.17.7 Final conclusion

The conclusion of a hypothesis test is always one of the following:

- reject the null hypothesis
- fail to reject the null hypothesis

The null hypothesis cannot be proven through a hypothesis test.

If H_0 is rejected, the conclusion should be worded as: > There is sufficient evidence to warrant rejection of the null hypothesis.

If H_0 is not rejected, the conclusion should be worded as: > There is insufficient evidence to warrant rejection of the null hypothesis.

1.17.8 Types of error

There are two types of error that can occur in hypothesis testing:

- **Type I error**
 - false positive
 - rejecting the null hypothesis when it is true
 - α represents the probability of a Type I error
- **Type II error**
 - false negative
 - failing to reject the null hypothesis when it is false
 - β represents the probability of a Type II error

Controlling errors

- For any fixed α , increasing the sample size n decreases β .
- For a fixed sample size n :
 - reducing α generally increases β , and vice versa.
- To reduce both α and β , increase the sample size.

1.18 Power of a hypothesis test

The power of a hypothesis test is the probability of correctly rejecting the null hypothesis when the alternative hypothesis is true.

It is given by:

$$\text{Power} = 1 - \beta$$